

## **Element A: Presentation and Justification of the Problem**

### **Problem Elaboration:**

While zoos are very important for education, conservation, and overall public support for wildlife, zoos also have plenty of issues. One of these issues include stereotypic behavior for various animals in zoos that derive from zoos' often small and disengaging enclosures.



In our case, a problem specifically arises in stereotypic behaviors and intellectual disengagement that take place in Amur Leopards at the Nashville Zoo which can have serious and detrimental effects on the leopard's health<sup>1</sup>. Stereotypic behaviors include walking in similar paths, having daily, unchanging, and disengaging activity, and an unchanging environment. These all contribute to a very boring and unhealthy landscape for the leopard.

### **Problem Definition:**

The problem is to create a device that can entertain, challenge, and engage the leopard enough to break the stereotypic environment of a zoo. While this can be done in an infinite amount of ways, the solution needs to be within budget, safe for the leopard to use, and easily operable by the zookeepers<sup>5</sup>.

## **Stakeholder Connections:**

### **Primary Stakeholders:**

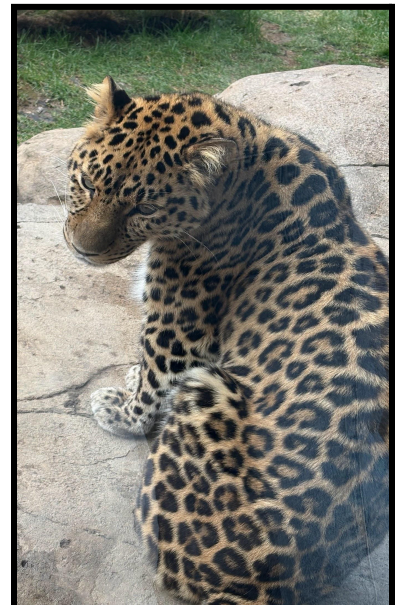
1. One of the zoo's head carnivore keepers is our main stakeholder, as she and other zookeepers will be setting up and cleaning the enrichment toy. They will also be the ones approving this project for usage in the habitat.
2. The leopard (specifically Natasha) will be utilizing our toy and the leopard's enrichment is the main goal of this project.

### **Secondary Stakeholders:**

1. Zoo guests who come to see the leopard may see the toy hanging in the exhibit or the leopard playing with the toy. An enriched leopard is a more healthy, interactive, and entertaining zoo exhibit that will raise more awareness for leopards.
2. Property managers or exhibit designers will have to compensate for the toy design or its effects on the leopard (i.e. different walking paths or increased behavior on trees).
3. Leopard Vets or general veterinarians that may work on the leopard. The toy will help to keep the leopard active and healthy.

## **Problem Justification:**

Since the leopard exhibit is rather new<sup>3</sup>, the leopards do not have an incredibly diverse selection of enrichment toys and this can lead to various mental and developmental issues. One of the main concerns for the leopards (or animals in general) is zoochosis which occurs during repetitive behaviors almost



exclusively found in captive animals. New enrichment toys that engage and break the leopards from their every day schedules keep this from happening. New toys are also great for development, conditioning, and overall health of the leopards<sup>4</sup>.

### **Citations:**

1\*-"What Is Zoochosis and How Do Animals Get It?" IDA USA, 2024,

[www.idausa.org/campaign/elephants/experts-agree-zoos-harm-good/what-is-zoochosis/](http://www.idausa.org/campaign/elephants/experts-agree-zoos-harm-good/what-is-zoochosis/).

2\*-Smithsonian's National Zoo & Conservation Biology Institute. "Animal Enrichment."

Smithsonian's National Zoo, 10 Oct. 2018, [nationalzoo.si.edu/animals/animal-enrichment](http://nationalzoo.si.edu/animals/animal-enrichment).

3\*-"Leopard Forest." Nashvillezoo.org, 2026, [www.nashvillezoo.org/leopard-forest](http://www.nashvillezoo.org/leopard-forest). Accessed 2 Apr. 2026.

4\*-Marketing Intern. "How Animal Enrichment Helps Wildlife Thrive in Human Care." Virginia Living Museum, 20 Nov. 2025,

[thevlm.org/how-animal-enrichment-helps-wildlife-thrive-in-human-care/](http://thevlm.org/how-animal-enrichment-helps-wildlife-thrive-in-human-care/).

5\*-Justification comes from our team meeting with the zookeeper. Note that responses were not documented directly, but paraphrased notes were taken among the team members.